

The Separator Certificate as One Loop: Rewrite, Bound, Subtract

Lecture notes, gcpo radius build

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Abstract

The separator certificate is one three-move loop, applied twice: *rewrite exactly* (law of total covariance), *bound only the remainder* (Cauchy–Schwarz), *get every standard deviation by subtraction* (law of total variance). The first pass conditions on the whole exterior and already yields a certificate; the second conditions on the hubs and yields the one the build uses. Three ledgers stay explicit throughout: which j ’s are computed and which are capped; where exactness ends; where each quantity comes from (Q side, Σ side, subtraction).

1 Two zones

A zero-mean Gaussian field $x \sim N(0, \Sigma)$, $Q = \Sigma^{-1}$ sparse; data point j carries $\eta_j = a_j^\top x$ with sparse support. The *group* of a target i is the top level sets of $|\text{Cor}(\eta_i, \eta_j)|$ over all j ; v is the deepest accepted absolute correlation (v_{last} in the build).

The build walks outward from $\text{supp}(a_i)$ on the graph of Q , avoiding the hubs H (few global latents: intercepts, small components), collecting a **core** K ; the **proposal** is $\text{lift}(K)$, the data points whose support touches K . Two zones, treated in opposite ways:

- **Inside the proposal:** finitely many candidates, each correlation computed exactly (sparse solves, no bounding). The group and v come from here.
- **Outside** (the *far field*: $\text{supp}(a_j) \cap K = \emptyset$, i.e. supported in $F := K^c$): never enumerated. The certificate is one j -free number bnd_i with $|\text{Cor}(\eta_i, \eta_j)| \leq \text{bnd}_i$ for every far-field j ; $\text{bnd}_i < v$ certifies the group complete.

Exactness is priced per j , and the far field is almost everybody. Hence the rule the rest of these notes keeps explicit: *everything j -dependent is computed inside the proposal, or capped j -free outside it.*

2 The loop

Normalize every variable to unit variance; covariances *are* correlations. One iteration:

Move 1 — rewrite exactly. Pick a conditioning set x_A ; the law of total covariance gives

$$\text{Cov}(\eta_i, \eta_j) = \underbrace{\text{Cov}(\mathbb{E}[\eta_i | x_A], \mathbb{E}[\eta_j | x_A])}_{\text{ledger term}} + \underbrace{\mathbb{E}[\text{Cov}(\eta_i, \eta_j | x_A)]}_{\text{remainder}}.$$

An identity: signed, nothing lost. A good A makes one term tractable — the remainder dies, or the ledger term becomes computable structure.

Move 2 — bound only the remainder. Cauchy–Schwarz, $|\text{Cov}(U, V)| \leq \text{sd}(U) \text{sd}(V)$, on the uncomputed term — never on the whole covariance. Every j -dependent factor then takes a j -free cap: a supremum computed once, or $\text{sd} \leq 1$.

Move 3 — sd by subtraction. The variables of Move 2 are global objects one cannot assemble; their *variances* survive, by the law of total variance:

$$\text{Var}(\mathbb{E}[\eta | x_A]) = \text{Var}(\eta) - \mathbb{E}[\text{Var}(\eta | x_A)] = \text{total} - \text{the computable part}.$$

For a GMRF the subtracted part is local: conditional on everything outside a set, the conditional precision is a subblock of Q — one small Cholesky. Compute what can be computed; subtract it from the total.

3 Level 0: conditioning on the whole exterior

Move 1. Condition on x_F . A far-field η_j is a function of x_F : the remainder vanishes and $\mathbb{E}[\eta_j | x_F] = \eta_j$, so

$$\text{Cov}(\eta_i, \eta_j) = \text{Cov}(\mathbb{E}[\eta_i | x_F], \eta_j) \quad \text{exactly.} \quad (1)$$

Only the part of the target *visible to the exterior* carries covariance out; the private fluctuation $\eta_i - \mathbb{E}[\eta_i | x_F]$ left the ledger by an identity, not an estimate.

Move 2. $|\text{Cov}(\eta_i, \eta_j)| \leq \text{sd}(\mathbb{E}[\eta_i | x_F]) \cdot 1$.

Move 3.

$$\text{Var}(\mathbb{E}[\eta_i | x_F]) = 1 - \sigma_i^2, \quad \sigma_i^2 = \mathbb{E} \text{Var}(\eta_i | x_F) = a_K^\top Q_{KK}^{-1} a_K :$$

given x_F only x_K is random, with conditional precision Q_{KK} .

Theorem 3.1 (Level-0 certificate). *For every far-field η_j , $|\text{Cor}(\eta_i, \eta_j)| \leq \sqrt{1 - \sigma_i^2}$.*

Why it is not enough. The visible variance contains everything the hubs explain: $1 - \sigma_i^2 \geq \rho_i^2$ (the hub-explained share), a floor at ρ_i . Correlations through a bottleneck *multiply*: with hub share t on both sides, true far-field correlations — and v with them — plateau near t , while the bound plateaus near $\sqrt{t}$. The bound is the square root of the thing it must beat; nothing certifies.

4 Level 1: conditioning on the hubs

Run the loop again on the surviving pair of (1), with conditioning set x_H .

Move 1. Total covariance plus the tower rule $E[E[\eta_i | x_F] | x_H] = E[\eta_i | x_H]$ give

$$\text{Cov}(\eta_i, \eta_j) = \underbrace{\text{Cov}(h_i, h_j)}_{\text{hub channel (ledger)}} + \underbrace{\text{Cov}(l_i, e_j)}_{\text{leak channel (remainder)}} \quad \text{exactly,} \quad (2)$$

with $h = E[\eta | x_H]$, $l_i = E[\eta_i | x_F] - E[\eta_i | x_H]$, $e_j = \eta_j - h_j$. The division of labour has flipped: at level 0 the remainder died and the ledger carried everything; here the ledger term is computable structure ($|H|$ -dimensional regressions) and the remainder is what Cauchy–Schwarz gets. Each pass peels one more computable layer off the unknown.

Move 2. Both channels are capped — the far field is never touched, so nothing stays unbounded:

$$|\text{Cov}(h_i, h_j)| \leq \rho_i \rho_j \leq \rho_i \rho_{\max}, \quad |\text{Cov}(l_i, e_j)| \leq \text{sd}(l_i) \text{sd}(e_j) \leq \text{sd}(l_i) \cdot 1.$$

Provenance of $\rho_{\max}$: the maximum of ρ_j over *all* data points. It is free — every j is the target of its own build, so the hub shares exist for the whole field anyway ($|H|$ global solves for the hub columns, then one $|H|$ -dimensional quadratic form per point) — and it is global rather than per-far-field because the far field varies with i ; one maximum dominates them all. The price is conservatism: the maximizer may sit inside i 's own proposal. (The per- j prohibition concerns per-*pair* work; a once-per- j vector amortized over all targets is exempt.)

Move 3. Two subtractions in cascade:

$$\text{Var}(l_i) = \underbrace{(1 - \sigma_i^2)}_{\text{level 0: total minus private}} - \underbrace{\rho_i^2}_{\text{level 1: minus the hubs}}, \quad \rho_i^2 = c_H^\top \Sigma_{HH}^{-1} c_H, \quad c_H = (\Sigma a_i)_H,$$

the latter on the Σ side from the shared hub columns. Each level subtracts the layer it just made computable; the rest is surrendered to Cauchy–Schwarz.

Theorem 4.1 (Level-1 certificate, as used). *For every far-field η_j ,*

$$|\text{Cor}(\eta_i, \eta_j)| \leq \text{bnd}_i := \rho_i \rho_{\max} + \sqrt{1 - \sigma_i^2 - \rho_i^2}.$$

$\text{bnd}_i < v$ (strictly, outside the tie band) certifies the group complete.

Remark 4.2 (R_i^2 is a repackaging). The companion notes write the leak term as $\sqrt{1 - \rho_i^2} \sqrt{R_i^2}$, $R_i^2 = 1 - \sigma_i^2 / (1 - \rho_i^2)$. Here it appears directly as the double subtraction $\sqrt{1 - \sigma_i^2 - \rho_i^2}$; R_i^2 is the same number normalized by the hub residual, and need never be introduced.

5 Why splitting before bounding never loses

Both levels are pure bounding; what did the second pass buy? A theorem:

Lemma 5.1 (Channelwise beats lumped). *For each far-field j ,*

$$\rho_i \rho_j + \text{sd}(l_i) \text{sd}(e_j) \leq \sqrt{\rho_i^2 + \text{Var}(l_i)} \sqrt{\rho_j^2 + \text{Var}(e_j)} = \sqrt{1 - \sigma_i^2} \cdot 1.$$

Proof. Cauchy–Schwarz in $\mathbb{R}^2$ on $(\rho_i, \text{sd}(l_i))$ and $(\rho_j, \text{sd}(e_j))$; the right side evaluates by $\rho_i^2 + \text{Var}(l_i) = 1 - \sigma_i^2$ and $\rho_j^2 + \text{Var}(e_j) = 1$. $\square$

Two readings of the gain. *Weaker worst case:* lumped Cauchy–Schwarz charges for “the whole visible part aligns with η_j ”; channelwise only for “hub aligns with hub, leak with residual”. The finer the rewrite, the less alignment information each bound discards. *The product cap:* after splitting, the hub cap is $\rho_i \rho_{\max}$, inheriting the multiply-through-a-bottleneck structure; lumped bounding buries ρ_i^2 under the root and floors at ρ_i .

One caveat: after uniformization the comparison stops being a theorem. At $\rho_{\max}$ near 1 the level-1 cap $\rho_i \rho_{\max} + \text{sd}(l_i)$ can exceed $\sqrt{1 - \sigma_i^2}$ (at $\rho_{\max} = 1$ it always does, slightly). That regime is the failure mode “ H too greedy”, where nothing certifies anyway; and both caps being valid, their minimum is free. (The build uses the level-1 cap alone.)

6 The bounding map

step	status
the two rewrites (1), (2)	exact, signed identities
in-proposal correlations	computed exactly; v comes from here
channelwise Cauchy–Schwarz	first loss; the sign of each channel is spent here (with the triangle inequality that adds the caps)
caps $\rho_j \leq \rho_{\max}$, $\text{sd}(e_j) \leq 1$	second loss; the price of one number for the whole far field
sealed core ($l_i = 0$)	no loss: $\text{Cov}(\eta_i, \eta_j) = \text{Cov}(h_i, h_j)$ exactly; per- j cost drops to an $ H $ -dimensional inner product and the build resolves the far field analytically, no certificate

Everything above the Cauchy–Schwarz row is signed and exact; opposite-signed channels cost tightness, never validity.

7 Cost, failure, level 2

Per target. One Cholesky of Q_{KK} (σ_i^2 , Q side); one $|H|$ -dimensional quadratic form (ρ_i^2 , Σ side, amortized); one subtraction; one comparison with v . Per far-field j : nothing.

Failure is a size, not an error. $\text{bnd}_i \geq v$ means K is too small. Growing K feeds σ_i^2 and starves $\sqrt{1 - \sigma_i^2 - \rho_i^2}$ monotonically — the mathematics of the grow-retry rounds. Smooth fields, or v at the noise floor, refuse and fall back to exact solves: conservative, never wrong.

Level 2. The loop does not care what the conditioning variables are. Hyperparameters θ in the slot ($\text{Cov} = \text{E}_\theta \text{Cov}_\theta + \text{Cov}$ of the per- θ means) would lift the certificate to mixture-level correlations; hard linear constraints already ride the slot as a low-rank correction. One more global mechanism, one more pass of *rewrite*, *bound*, *subtract*. The loop is the theory.